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APPENDIX

A LIMITATIONS

While this paper provides valuable insights into how large-scale pretraining corpus contributes to the emergent abilities of LLMs through n -gram search, there are a few limitations that we want to list out. First, the model we use, Pythia (Biderman et al., 2023), and the pretraining corpus, Pile (Gao et al., 2020), are slightly outdated and have been outperformed by many new open-source LLMs. However, most open-source LLMs lack corresponding pretraining data and have limited model sizes and pre-training checkpoints, hindering scaling effect studies. The current WIMBD system also has limitations in searching larger corpora like Dolma (Soldaini et al., 2024a) (3T tokens), calling for improved searching and retrieval methods. The quality of task-relevant n -gram pairs is highly sensitive to the filtering method, and while the current embedding similarity-based approach is effective, better filtering methods could significantly enhance the analysis, which is left for future research.

B ETHICS STATEMENT

The insights and methodologies developed in this paper have several significant implications for the broader field of artificial intelligence, particularly in the development and deployment of large language models (LLMs). Understanding the balance between memorization and generalization within LLMs is crucial for both advancing the theoretical foundation of machine learning and addressing practical concerns related to their use.

Enhanced Model Interpretability: By extending the definition of memorization and examining how LLMs utilize their pretraining data, our research contributes to a deeper understanding of the internal mechanics of these models. This improved interpretability can help researchers and practitioners diagnose and mitigate issues related to data bias, model robustness, and unexpected behaviors in AI systems.

Privacy and Security Considerations: Our findings have direct implications for privacy and security in AI. Demonstrating how LLMs memorize and potentially recall training data underscores the need for rigorous data handling and anonymization techniques. It raises awareness about the risks of inadvertent leakage of sensitive information, thereby informing policy and best practices for data usage in training large models.

Economic and Societal Impact: As LLMs become more integral to various industries, understanding their capabilities and limitations can have significant economic and societal implications. Our research can help businesses and policymakers make informed decisions about deploying these models, ensuring they are used ethically and effectively. This, in turn, can lead to more reliable and trustworthy AI systems, fostering greater public trust and acceptance.

C RELATED WORK DETAILS

Understanding LLMs’ capabilities from training data Prystawski et al. (2023) and Wang et al. (2024) discuss how the reasoning ability of language models is a consequence of their pretraining data. Prystawski et al. (2023) discusses how chain-of-thought reasoning is effective in autoregressive language models because of local structure within pretraining data, and Wang et al. (2024) derives novel conclusions from known facts by aggregating reasoning paths seen in pretraining data. On the other hand, Xie et al. (2022) and Wang et al. (2023) discuss how in-context learning is a by-product of learning the pretraining data distribution. They both suggest that language models learn to implicitly infer a latent variable from the given prompt, as the pretraining data is generated from some unknown latent variable. Additionally, Chan et al. (2022) propose that the distributional properties of training data drive emergent in-context learning behaviors in large language models, whereas Razeghi et al. (2023) shows the influence the pretraining data on the mathematical abilities of LLMs. Chen et al. (2024) also highlights the significance of parallel structures in pretraining data for the emergence of in-context learning.

However, the small-scale nature of such analysis is antithetical to the commonly believed main driving factor behind the performance of LLMs: scaling. Recently, Kirchenbauer et al. (2024) proposes to provide statistical evidence of the dependence of a target model capabilities on subsets of its training data, by estimating the data distribution with an embedding-induced kernel. However, their estimation is based on a very small portion of the pretraining data (around 0.3%) as computing the embeddings of a huge dataset is very non-trivial. To get a better estimation of the whole distribution of the pretraining data, Elazar et al. (2024) construct a retrieval system, **WIMBD**, that can efficiently search n -gram phrases over hundreds and thousands of GBs of pretraining data. Merrill et al. (2024) propose an efficient data structure that enables unbounded-length n -gram searches in massive pretraining datasets, and find that larger LLMs generate less novel large n -grams compared to human written texts, and (Shaib et al., 2024) shows how syntactic templates from the training data causes models to re-use such templates after training. Liu et al. (2024) proposes to build a prefix-based efficient retrieval system for pretraining corpora, and then construct large-scale ∞ -gram language models to estimate the distribution of text corpora. However, such distribution only models local contextual dependency, which might not be useful for understanding complex LLM capabilities.

New methods and analysis to investigate these capabilities at scale and to understand the role of scaling are needed to obtain useful insights into real-world LLMs. In this work, we aim to provide an in-depth analysis of the origin of the general zero-shot capabilities of LLMs, by performing full searches across the whole pretraining corpus with the WIMBD and ∞ -gram framework.

Memorization v.s. generalization The phenomenon of machine learning models being able to perfectly memorize the training data has been studied in many previous works. Most of them define LLM memorization as exactly recalling the training examples by designed prompting, including the memorization of rare long-tail data, like private information (Zhang et al., 2023), and the contamination of testing sets (Jiang et al., 2024). Carlini et al. (2022) found that the exact copy and pasting behaviors are more prevalent in larger LMs. Hartmann et al. (2023) provides a comprehensive summarization of LLM memorization.

Several papers have studied the interplay between memorization and generalization of training data. Feldman (2020) prove that memorizing the training data is in fact required for optimal generalization on testing data. Works along this line (Feldman and Zhang, 2020; Zhang et al., 2023) extend the original definition of memorization by quantifying the extent of memorizing a training example with the performance difference when including and excluding this specific example in training data. However, this definition is impractical for large-scale analysis of LLMs as it requires retraining an LLM from scratch to analyze one data point. In this paper, we propose a new definition of distributional memorization by using n -gram counts, which is more suitable for large-scale analysis with LLMs.

D EXPERIMENT DETAILS

Dataset choice The choice of dataset comes from the consideration of balancing different types of tasks, and we believe the combination of translation + factual QA + reasoning can well represent the spectrum of possible tasks. To enhance the selection of reasoning tasks, we also add GSM8K as a new task as described in point 1 of the general response. We found both the performance v.s. n -gram count results and the distributional generalization results on GSM8K align with our previous findings.

Embedding model choice The choice of embedding model comes from accommodating the need for different tasks. We use the LASER embedding model for the Translation task because it is specialized for translation. We use E5 for other tasks because it is a general-purpose sentence embedding model that performs cosine-similarity-based retrieval.

Task-gram table construction To build a task-gram table, we use a two-step process. First, we filter all possible input-output n -gram pairs in a supervised dataset by the cosine similarity between their embeddings. Second, we search through the pretraining corpus and keep n -gram pairs that have nonzero counts.

For WMT, we use the Europarl (Koehn, 2005) parallel corpus to construct a large task-gram table. For TriviaQA, we use its training set, while for MMLU, we use the testing set as there is no training set available. We employ two different models to filter the n -grams pairs. For translation tasks, we